Optimization of Capacity Allocation Using Linear Integer Programming in a Semiconductor Manufacturing Company

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ABSTRACT

This research addresses the optimization of capacity allocation in the photolithography area in a semiconductor manufacturing company using Linear Integer Programming (LIP). The primary objective of this research is to optimize machine utilization rates and balance workloads across all machines while meeting wafer demand. To address this problem, we aim to develop an LIP model that optimizes capacity allocation and determines the optimum wafer allocation quantities for each machine and recipe. The study begins with a comprehensive explanation of semiconductor manufacturing, emphasizing its significance and challenges in the photolithography area. The methodology includes problem identification, data collection, analysis, LIP model development to be applied using MATLAB, and results validation through sensitivity analysis. Results show that the optimization improved the average machine utilization from 88.87% to 69.46%, and reduced the Mean Absolute Deviation (MAD) from 43.23 to 0.003, indicating a more balanced workload. The study concludes with key findings, contributions to semiconductor manufacturing field, and directions for future research.

INTRODUCTION

Semiconductor manufacturing, especially photolithography process, is often knows for its high-complex and capital-intensive industry that also involves complex processes from transferring circuit patterns into wafers using expensive machines such as steppers and tracks. With this condition and the critical part of photolithography, it is important to optimize capacity allocation to improve operational efficiency and reduce overall cycle times. This research aims to develop a Linear Integer Programming (LIP) model to optimize machine utilization rates, balance machine workloads across all machines, while also meet wafer demand without overutilizing any of these machines. With this optimized capacity allocation, it will improve the production efficiency and help the company in the decision-making process.

LITERATURE REVIEW

Previous research on capacity allocation problem has utilized various methods and computer tools to solve this problem with different conditions. Ghasemi et al. (2020) proposed a Mixed Integer Linear Programming (MILP) combined with an improved genetic algorithm to solve capacity allocation and scheduling issues. Chen et al. (2016) used a genetic algorithm with an integer programming model, while Toktay and Uzsoy (1996) used a maximum flow problem approach with integer-side constraints. Klemmt et al. (2010) employed a multistage MILP model to do production planning in the photolithography area. These researches gave valuable knowledge but also had room for improvements.

METHODOLOGY

The methodology for this research is started with the problem identification from the collected data from a semiconductor manufacturing company, with data of machine process capabilities, wafer demand, tool processing times, and machine availability. These data then analyzed in order to develop the right LIP model to minimize the maximum total machine utilization rates of all machines for all recipes with wafer allocation as the decision variables. The model is coded and solved using Optimization Toolbox in MATLAB, and the validation process are done through sensitivity analysis method to ensure the results accuracy and model's reliability.

MODEL DEVELOPMENT

The LIP model is represented as:

$$\textbf{Minimize } Z = Maximum\left(\sum_{j=1}^{r} U_{1j}, \sum_{j=1}^{r} U_{2j}, \sum_{j=1}^{r} U_{3j}, \sum_{j=1}^{r} U_{4j}, \sum_{j=1}^{r} U_{5j}, \sum_{j=1}^{r} U_{6j}, \sum_{j=1}^{r} U_{7j}, \sum_{j=1}^{r} U_{8j}, \sum_{j=1}^{r} U_{9j}, \sum_{j=1}^{r} U_{10j}, \sum_{j=1}^{r} U_{11j}\right) \quad (1)$$

Subject to

$$\sum_{j=1}^{r} U_{ij} \leq 100\% \quad \text{, for all } I \quad (2)$$

$$\sum_{i=1}^{n} x_{ij} = D_j \quad \text{, for all } j \quad (3)$$

$$mpc_{ij} = [0,1] \quad (4)$$

$$x_{ij} \in Z^{+} \quad \text{, for all } i \text{ and } j \quad (5)$$

$$x_{ij} \geq 0 \quad \text{, for all } i \text{ and } j \quad (6)$$

Where x_{ij} represents represent the number of wafers allocated to machine i for recipe j, $U_{ij} = \frac{x_{ij} * p_{ij}}{t_{ij}} * 100\%$ and it represents the utilization rate of machine i for recipe j, p_{ij} is the value of tool processing time of machine i for recipe j, mpc_{ij} represents the machine process capability constraint of machine i for recipe j, t_{ij} is the value of machine availability of machine i for recipe j, n is the number of machines, r is the number of recipes, and D_j represent the demand for each recipe.

RESULTS AND DISCUSSION

In the current practice, the average machine utilization is 88.87%, showing that these machines are operating near or over their capacity (as shown in Figure 1). This high utilization rate may lead to operational inefficiencies and potential bottlenecks in the manufacturing process. Moreover, the Mean Absolute Deviation (MAD) of machine utilization rate in current practice is 43.23. This is a relatively high number and this significant deviation means that there are some machines that are being overutilized while others are underutilized.

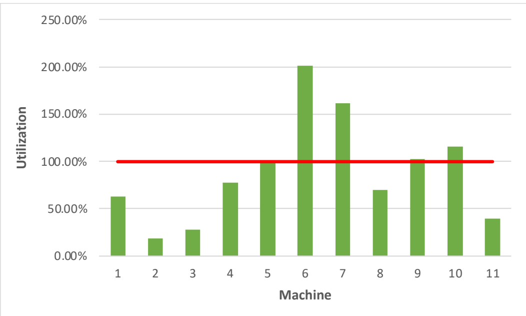

Figure 1: *Current Utilization of Each Machine*

After applying the LIP model, there is a huge improvement in the average machine utilization rate, declining to 69.46%. This improvement implicates that there is a more balanced and efficient used of the machines (as shown in Figure 2), switching wafer loads from overutilized machines to underutilized machines. Furthermore, the MAD value of machine utilization was heavily reduced to just 0.003, showing a more uniform distribution of workloads across all machines, thus will be important for keeping a steady production flow and enhancing the overall operational efficiency.

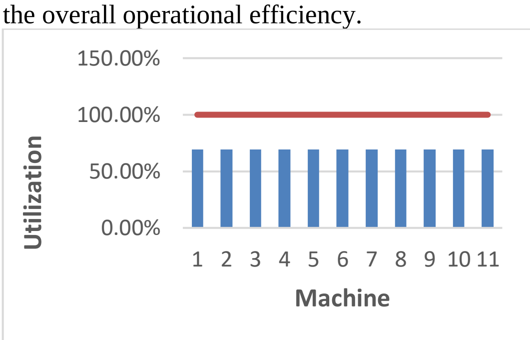

Figure 2: *Optimized Utilization of Each Machine*

CONCLUSION

The research has successfully developed and applied a LIP model to optimize capacity allocation in a semiconductor manufacturing company and determined the optimum wafer allocation quantities for each machine of each recipe. The LIP model has improved machine utilization rates and balanced distribution of machine workloads, contributing to both theoretical knowledge and practical applications in this field. Future research can explore further improvements to the model and its applicability to other manufacturing process in semiconductor manufacturing or even other industries.

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